

frequency recovery in all-digital modems for burst- mode transmissions," Communications, IEEE Transactions on , vol.43, no.2/3/4, pp.1 169, 1178, Feb./March/April 1995, which is hereby incorporated by reference, may be employed to this purpose. This step is performed in the frequency/phase tracking unit 256.

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Channel estimation is performed on the basis of the non-precoded or precoded pilots (i.e. the one or more second fields) for each waveform component, wherein the estimated channel is to be sent back to the gateway only if it is deemed to be reliable, as it has been described in the context of the previous embodiments.

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It should be noted that features described for a specific embodiment may be combined with the features of other embodiments, unless the contrary is explicitly mentioned. Based on the description and figures that have been provided, a person with ordinary skills in the art will be able to construct a computer program for implementing the described method steps without undue burden.

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This work has been partially supported by European Space Agency, ESA, funded project OPTIMUS ("Optimized transmission Techniques for satcoM UnicaSt Interactive traffic"), contract 40001 1642 1/1 5/NL/FE.

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It should be understood that the detailed description of specific preferred embodiments is given by way of illustration only, since various changes and modifications within the scope of the invention will be apparent to the person skilled in the art. The scope of protection is defined by the following set of claims.